Independent local validation of the VigilODE PR90–94 integration

VigilODE validation team
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We replayed source-sensitive support checks at their original commits and ran a separate native validation on the integrated delivery plus limited contract repairs. The original 23-node, 34-edge plan, 73-path coverage, ten interactions and negative evidence remain intact. The native contract and manufactured-model results below are scoped; full research-reference authority, public API compatibility and broad physical-client qualification remain open.

I. IDENTITY AND METHODS

Delivered source commit:
`e935878fefe795a1b7e88aba64e5e30ef7be1578`
Delivered source tree:
`bff4df3e8f2be76dccfb072302556d4fea4b5dc2`
Native validation code commit:
`5c57bf146f2f5faf69f711f133a83fcb265cee90`
Native validation code tree:
`9633bde7dda1e9dd4748d1aecc8f22af68b5f5ac`

The pure preregistration commit was remote-reachable before execution, but the code execution commit was local when measurements began; this process deviation is retained rather than retroactively passed. The validation used locked offline Rust 1.94.1 dependencies, one Cargo build and one timing process at a time, single-thread solver settings and a pinned CPU for the short matched-accuracy cohort. Raw receipts and executable hashes are indexed in `EVIDENCE_MANIFEST.json`. Source-locked exact checks are not counted as current native or physical accuracy checks.

II. NATIVE AND PHYSICAL RESULTS

TABLE I. Final-tree contract execution. Ignored tests are not counted as passes.

Scope	Passed	Ignored	Outcome
Facade and coupled measurement suites	424	0	Exit 0
Core/facade/coupled release command	830	21	Exit 0
Integrator release command	1196	54	Initial 1800-s timeout retained; bounded second attempt exit 0
Native VODE/CVODE/diffsol adapter grid	7	0	Common linear exact solution and endpoint parity; exit 0

The release smoke returned all sixteen state components for four manufactured linear-decay paths. The largest returned-state error was 3.3719×10^{-9}; RHS-only paths made zero JVP calls. These are manufactured oracles, not validation of an external physical client. The inherited W-filtered Prothero–Robinson negative control still fails its cost gate: accepted-step ratio $15/7 = 2.143 > 1.3$. The Off arm has interior error 311.467. An expected-negative reproduction does not make that route productive.

III. MATCHED ACCURACY AND PROFILE

For the rapid oscillator, both retained DOPRI and RHS-only switching produced the same 17-point maximum state error 9.4603×10^{-7} at input tolerance 10^{-8}, below the common achieved-error cap 10^{-6}. Their maximum phase and amplitude errors were 1.4996×10^{-7} radians and 9.8307×10^{-7}. They used identical work counters: 10027 accepted steps, 10845 attempts and 65071 RHS calls, with zero JVP calls. Three timed samples gave median solver-reported times 5.591 ms (DOPRI) and 5.702 ms (switching). The small difference is diagnostic only. At input tolerance 10^{-6} both arms exceeded the common 10^{-4} achieved-error cap, so no speed ratio was assigned to that first cohort. Separate Callgrind differences between 16 and 32 windows (17 and 33 requested observation points) at the same endpoint gave

14827 and 14849 marginal guest instructions per added point, respectively. Numerical work changed between grids, so these are neither whole-solve instruction counts nor native elapsed time.

IV. OPEN GATES AND CLAIM CEILING

The full research-reference authority check fails on this checkout with 49 missing citations; the PR94 CI failure reported 53 under its own snapshot. The attempted Q-LEDGER repair and a proposed stiff-cubic benchmark driver were rejected in independent review and were not merged. The standalone vendor validator was invoked without its required directory argument (exit 2), leaving that graph validation unperformed, although locked offline metadata and builds passed. Whole-workspace format checking still fails on the same two coupled files as the delivered source. Public switching configuration changes still need an explicit compatibility or version decision. The ten prospective interaction campaigns, six candidate implementation nodes and real-client qualification are not promoted by the tests above. Broad production remains **HOLD**.